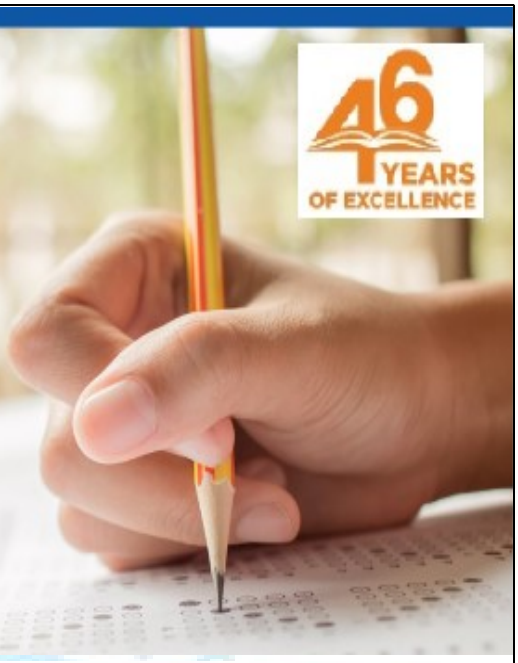
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 29-03-26**
Time: 3 Hrs **Max. Marks: 198**

29-03-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-2(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 198

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				**18**	**66**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				**18**	**66**

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MATHEMATICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let $P(x)=x^2+ax+b$ and $Q(x)=x^2+cx+d$ be quadratic polynomials with real coefficients. If $\tan\theta_1,\tan\theta_2$ are the roots of $P(x)$ and $\cot\theta_1,\cot\theta_2$ are the roots of $Q(x)$ for some $\theta_1,\theta_2\in(0,\pi/2)$, such that $\theta_1+\theta_2=\pi/4$ and $P(1)\cdot Q(1)=16$, then the value of $\frac{a+c}{1+d}$ is equal to:

A) 1 B) 2 C) 3 D) 4

2. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=x+|x|\cos x$, then which of the following statements is **TRUE**

A) f is one-one, but NOT onto B) f is onto, but NOT one-one

C) f is BOTH one-one and onto D) f is NEITHER one-one nor onto

3. Let the functions $f:\mathbb{R}\to\mathbb{R}$ *and* $g:[-3,3]\to\mathbb{R}$ be defined by:

$f(x)=\min\{|x|+1,3-|x|\}$

$g(x)=\sqrt{9-x^2}$

If S is the region in the $xy-$plane defined by $S=\{(x,y):f(x)\le y\le g(x)\}$, then the area of the region S is

A) $\frac{9\pi}{2}-6$ B) $\frac{9\pi}{4}-\frac{7}{2}$ C) $\frac{9\pi}{2}-7$ D) $9\pi-14$

4. Let a,b and k be positive real numbers. Suppose Q is a point on the parabola $y^2=4kx$ such that its distance from the focus is $2k$ and it lies in the first quadrant. An ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through Q. If the normal to the parabola at Q and the tangent to the ellipse at Q are parallel, then the eccentricity of the ellipse is

A) $\sqrt{\frac{2}{3}}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{1}{\sqrt{2}}$ D) $\frac{1}{\sqrt{3}}$

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5. Let S be the set of all possible values of the determinant of a matrix $M=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a,b,c,d are chosen independently and at random from the set $\{0, 1\}$. Let X be a random variable representing the value of $\det(M)$. Now, consider a quadratic equation $t^2-\gamma t+\delta=0$, where γ and δ are two values chosen independently from the set S with probabilities proportional to their frequency of occurrence in the experiment of forming matrix M. The probability that the roots of this quadratic equation are non-real complex numbers is

A) $\frac{31}{64}$ B) $\frac{37}{128}$ C) $\frac{19}{64}$ D) $\frac{48}{256}$

6. Consider the region $R=\left\{(x,y)\in\mathbb{R}^2 : 0\le x\le\frac{\pi}{3}, 0\le y\le 4\cos\left(3x-\frac{\pi}{2}\right)\right\}$. A rectangle is inscribed in R with one side on the x-axis. Let A_0 be the area of the rectangle that has the maximum perimeter among all such rectangles, then the value of $\frac{9A_0}{\sqrt{35}\sin^{-1}(1/6)}$

A) 4 B) 3 C) $\frac{1}{4}$ D) None of above

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

7. Let $I=\int\frac{(\sin^3\theta-\cos^3\theta-\cos^2\theta)(\sin\theta+\cos\theta+\cos^2\theta)^{2010}}{(\sin\theta)^{2012}(\cos\theta)^{2012}}d\theta=\frac{(f(\theta))^p}{2011}+C$ then identify the correct statement(s)

A) $(f(\pi/3),p)=(2+\sqrt{3},2012)$ B) $(f(\pi/3),p)=(2-\sqrt{3},2011)$

C) $(f(\pi/3),p)=(2+\sqrt{3},2011)$ D) $(f(\pi/4),p)=(1+\sqrt{8},2011)$

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8. Let A be a 3×3 non-singular matrix with real entries such that $A^2 = A^T$, where A^T is the transpose of A. *If* $A \neq I$, then which of the following statements is/are **TRUE**

A) $\det(A) = 1$ B) $\text{tr}(A) = 0$ C) $A^2 + A + I = O$ D) $\det(A^2 + A + I) = 0$

9. Let z be a **complex number** such that the following two conditions are met:

1. The distance between points z and 1 in the **complex plane** is 2.

2. The distance between points z^2 and 1 in the **complex plane** is 6.

Let $\text{Re}(z)$ and $\text{Im}(z)$ represent the real and imaginary parts of z, respectively, then which of the following statements is/are correct?

A) The value of $\left|\frac{\text{Im}(z)}{\text{Re}(z)}\right| \cdot \sqrt{7}$ *is* 4.2.

B) The real part of z is 1.25.

C) The square of the imaginary part of z is $\frac{63}{16}$.

D) The modulus of z is $\sqrt{7}$.

10. Let a_n be the number of subsets of $\{1,2,3,\ldots,n\}$ that do not contain any two consecutive numbers. Which of the following statements is/are correct?

A) $a_n = a_{n-1} + a_{n-2}$ *for* $n \geq 3$ B) $a_5 = 13$ C) $a_6 = 21$ D) $a_4 = 7$

11. Consider a plane P_1 that passes through the line of intersection of the two planes:

$$\begin{cases} x - y - z - 4 = 0 \\ x + y + 2z - 4 = 0 \end{cases}$$

The plane P_1 is parallel to the line of intersection of another pair of planes: $\begin{cases} 2x + 3y + z = 1 \\ x + 3y + 2z = 2 \end{cases}$

Let the equation of P_1 be expressed in the form $x + \alpha y + \beta z + \gamma = 0$.

Consider four parallel lines $L_1, L_2, L_3,$ and L_4 defined as:

$L_1 : \frac{x-3}{2} = \frac{y-3}{1} = \frac{z-3}{2}$

$L_2 : \frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{2}$

$L_3 : \frac{x}{2} = \frac{y-3}{1} = \frac{z}{2}$

$L_4 : \frac{x}{2} = \frac{y-3}{1} = \frac{z-3}{2}$

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A second plane P_2 intersects these four lines at points $V_1, V_2, V_3,$ and V_4 respectively. Let S be the minimum possible area of the quadrilateral $V_1V_2V_3V_4$ formed by these intersection points.

Based on the above information, determine which of the following statements is/are correct:

A) The value of $|\alpha+\beta+\gamma|$ is 11.

B) The minimum area S of the quadrilateral $V_1V_2V_3V_4$ is 5.

C) The vector normal to plane P_1 is parallel to $2\hat{i}-\hat{j}-3\hat{k}$

D) The sum $|\alpha+\beta+\gamma|+S$ is equal to **14**.

12. If $\int_0^1 \frac{\ln(x)}{1-x^2}dx = -\frac{\pi^2}{\lambda}$ *and* $\int_0^1 \ln(x)\cdot\ln(1-x)dx = \alpha - \frac{\pi^2}{\beta}$, then which of the following options is/are correct? (Given: $\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$)

A) $\alpha+\beta+\lambda$ is a perfect square.

B) $\alpha+\lambda=\beta$

C) $\alpha+\beta=\lambda$

D) $\alpha+\lambda$ is a perfect square.

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

13. Let $y=f(x)$ be a differentiable function for $x>0$ satisfying the differential equation:

$$x\frac{dy}{dx}+y=\frac{y^2\ln x}{x}$$

Given that the curve passes through the point $(1,1)$

Let $I=\int_1^e \frac{1}{x^2 f(x)}dx$

then the value of $4e^2(1-I)$ is equal to

For More Join: @IITJEE_Advanced

14. Let $\{a_n\}_{n\geq 1}$ be an arithmetic progression with first term $a_1 = 1$ and common difference $d > 0$. Suppose that a_2, a_5, a_{14} are three consecutive terms of a geometric progression. Let S_n denote the sum of the first n terms of $\{a_n\}$. Further, let $\{b_k\}_{k\geq 1}$ be a geometric progression with $b_1 = 1$ and common ratio equal to the common ratio of the progression formed by a_2, a_5, a_{14}. Let T_m denote the sum of the first m terms of $\{b_k\}_{k\geq 1}$, then the number of pairs of positive integers (n, m) with $n > 1$ such that $S_n = T_m$ is

15. Let $E = \dfrac{2\sin 2^\circ + 4\sin 4^\circ + \cdots + 178\sin 178^\circ}{\cot 1^\circ}$ then the value of E is

16. In a triangle ABC, let $\vec{u} = \overrightarrow{BC}, \vec{v} = \overrightarrow{CA}$, and $\vec{w} = \overrightarrow{AB}$. Given $|\vec{u}| = 4, |\vec{v}| = 6$, and

$$\frac{\vec{u}.\left(\vec{w} - \vec{v}\right)}{\vec{w}.\left(\vec{u} - \vec{v}\right)} = \frac{2}{5}$$

Find the value of $\left|\vec{u} \times \vec{v}\right|^2$

17. Number of integral values of "x" for which $(\log_{\{x\}}[x])^2 - 3\log_{\{x\}}[x] + 2 \leq 0$ is λ_1 & number of integral values of "x" for which $(\log_{[x]}\{x\})^2 - 3\log_{[x]}\{x\} + 2 = 0$ is λ_2 then $\lambda_1 + \lambda_2 = \ldots\ldots\ldots\ldots$ ($[.]$ *denotes G.I.F and* $\{.\}$ *denotes fractional part*)

18. The value of the integer k such that the following limit L is finite and non – zero,

$$L = \lim_{x\to 0^+} \frac{\ln(x^x)\cdot(e^{x^x - 1} - 1) - (x^x - 1)}{\sin^k(x\ln x)} \text{ is}$$

For More Join: @IITJEE_Advanced

PHYSICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

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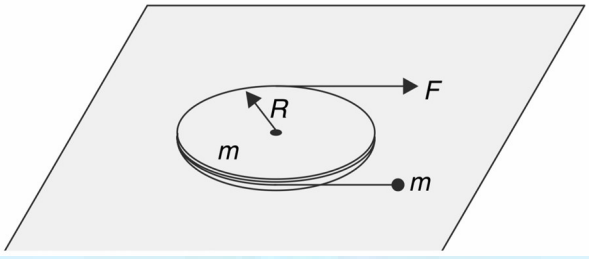
19. A particle undergoing circular motion on a fixed circle experiences a central attractive force towards a point A on the circular path. The attractive force F is inversely propontional to r^n where r is the distance between the particle and A.What can be the value of n.

A) –2 B) 3 C) 5 D) 1

20. The Thapi Method used by Indain masons measures the volume of sand by heaping it into a conical pile and measuring its circumference C. If the coefficient of friction between the sand panticles is $\tan^{-1}\left(\frac{\pi^2}{15}\right)$ then find the volume of sand.

A) $\frac{C^3}{360}$ B) $\frac{C^3}{120}$ C) $\frac{\pi C^3}{720}$ D) $\frac{\pi C^3}{120}$

21. A disc of mass m and radius R lies flat on a smooth horizontal table. A massless string runs halfway around it as shown in figure. One end of the string is attached to a small body of mass m and the other end is being pulled with a force F. The circumference of the disc is sufficiently rough so that the string does not slip over it. Find acceleration of the small body.

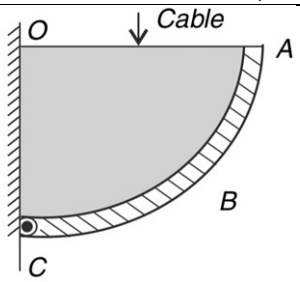


A) $\frac{F}{m}$ B) $\frac{F}{2m}$ C) $\frac{F}{3m}$ D) $\frac{F}{4m}$

22. A plate is in the shape of a quarter cylinder of radius R and length L. This plate is hinged at C to a vertical wall and can rotate freely about C. The end A of the plate is tied to the wall using two horizontal cables [the other cable is parallel to OA and the two cables are placed symmetrically. The space between the wall and the plate is filled completely with water (density = ρ). Neglect the weight of the plate and calculate the tension in each cable.

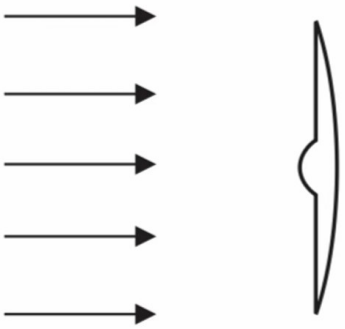
Take $\tan^{-1}\left(\frac{\pi}{2}\right) \simeq 57^0$ and $\cos 57^0 \simeq \frac{1}{2}$

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A) $\frac{\rho g R^2 L\sqrt{\pi^2+4}}{4}$ B) $\frac{\rho g R^2 L}{2}$ C) $\frac{\rho g R^2 L\sqrt{4+x^2}}{8}$ D) $\frac{\rho g R^2 Lx}{8}$

23.



A composite lens is placed coaxially in the path of a cylindrical light beam of diameter 5 cm as shown in the figure. The composite lens is made by gluing two plano-convex lenses. Diameters of the first and second lens are 1.0 cm and 5.0 cm and their focal lengths are 10 cm and 20 cm respectively. Where to the right of the lens a screen should be placed to obtain a light spot of smallest diameter on it?

A) 16 cm B) 18 cm C) 17.14 cm D) 15 cm

24. A region of space contains a uniform magnetic field $\vec{B}$ confined to a cylindrical volume of radius R. The magnetic field is increasing at a constant rate $dB/dt = \alpha$. A thin, non-conducting circular ring of radius $r < R$ is placed coaxially within this field. Two small, identical metallic beads are fixed at point F and Q on the ring, such that they subtend an angle at 120^0 at the center O.

A student is asked to determine the potential difference between points F and Q along the shorter arc of the ring. He now connects a high-resistance ideal voltmeter between point F and Q. The voltmeter is connected used three different wire configurations:

1. Placement A : The voltmeter leads follow the shorter arc of the ring from F to Q.

2. Placement B : The voltmeter leads follow the longer arc of the ring from F and Q.

3. Placement C : The voltmeter is placed at the center O, and the leads follow the radii OP and OQ.

Mark the correct statement(s).

For More Join: @IITJEE_Advanced

A) The potential difference $V_P - V_Q$ is exactly $\frac{1}{6}\pi r^2\alpha$, because the induced electromotive force (EMF) is distributed uniformly along the circular path of the electric field lines.

B) The voltmeter reads zero in all cases because F and Q are on a non-conducting ring and no current flows.

C) The voltmeter reads $\frac{1}{3}\pi r^2\alpha$ in all cases because it measures the EMF between the two points regardless of the lead path.

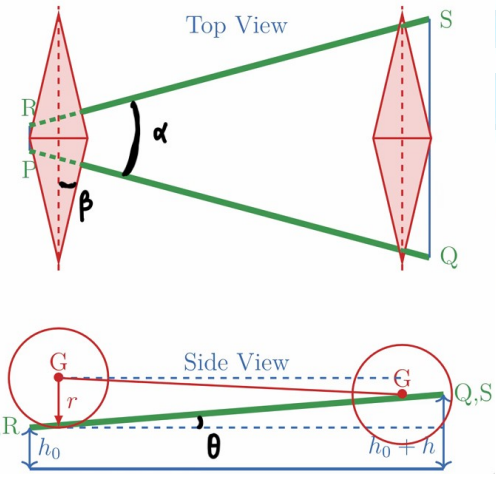
D) The voltmeter reads $\frac{1}{3}\pi r^2\alpha$ for Placement A, $\frac{2}{3}\pi r^2\alpha$ for placement B, and zero for placement C.

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

25. An uphill roller is a uniform double cone made by joining 2 identical cones at the base. This double cone (COM at G) is placed on 2 rods (PQ & RS) that are inclined with each other at an angle α and make an angle θ with the horizontal (as shown in the diagram)



For certain values of β (semi-vertical angle of cone) the double cone seems to climb up the incline since G moves down in the process (going from PR to QS).

Mark the correct statement(s):

For More Join: @IITJEE_Advanced

A) For the double cone to 'roll up' the incline, the semi-vertical angle must follow

$\tan\beta \tan\left(\frac{\alpha}{2}\right) > \sin\theta$

B) For the double cone to 'roll up' the incline, the semi-vertical angle must follow

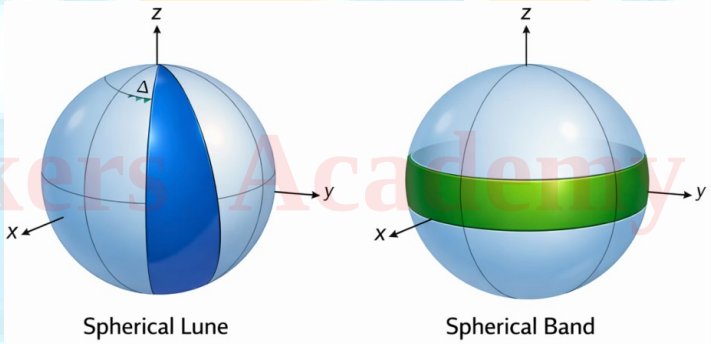
$\tan\beta > \tan\left(\frac{\alpha}{2}\right).\sin\theta$

C) For the double cone to 'roll up' the incline, the semi-vertical angle must follow

$\sin\beta > \sin\left(\frac{\alpha}{2}\right).\tan\theta$

D) The above phenomenon occurs at the same value(s) of β for both a double solid cone & a double hollow cone.

26. The lines moving from pole to pole of a spherical shell are known as longitudes whereas the lines moving perpendicular to the axis of a shell are known as latitudes. A spherical lune is the section between two longitudes whereas a spherical band is the section between two latitudes as shown below.



Mark the correct statement(s).

A) The COM of a spherical lune between 2 longitudes seperated by very small angle θ will be identical to that of a semi-circular ring.

B) The COM of a spherical lune between 2 longitudes seperated by angle θ_0 is at a distance $\frac{\pi R}{2\theta_0}\sin\left(\frac{\theta_0}{2}\right)$ from geometric centre.

C) The COM of a spherical lune between 2 longitudes seperated by angle θ_0 is at a distance $\frac{2R}{\pi\theta_0}\sin(\theta_0)$ from geometric centre

D) The COM of a spherical ring between 2 latitudes separated by very small angle θ will be identical to that of a circular ring.

For More Join: @IITJEE_Advanced

27. Two spherical blackbody stars, Star A and Star B, are observed by an astrophysicist. The following data is collected:

1. The wavelength at which the spectral emissive power is maximum for star A is $\lambda_A = 400\,nm$.

2. The wavelength at which the spectral emissive power is maximum for star B is $\lambda_B = 800\,nm$.

3. The total radiant power (Luminosity) measured from both stars is found to be identical $(P_A = P_B)$.

If R_A and R_B are the radii of Star A and Star B respectively,

Mark the correct statement(s):

A) The ratio of the surface areas of the stars is $S_A : S_B = 1:16$.

B) If both stars are observed through a narrow-band filter centered at 10 nm (X-ray region), the spectral power measured from Star A will be significantly higher than that from Star B.

C) The maximum value of the spectral emissive power $(E_\lambda)_{\max}$ for star A is 32 times that of star B.

D) If the distance of star A from Earth is twice the distance of Star B, the apparent brightness (flux) of both stars as seen from Earth will be the same.

28. A group of Olympiad students is attempting to determine the molar gas constant R by measuring the pressure P, volume V, and temperature T of $n = 1.00$ mole of an ideal gas. The following measurements and their absolute uncertainties are recorded:

Volume (V) : $20.00 \pm 0.02L$

Temperature (T) : $300.00 \pm 0.3K$

Pressure (P) : The students measure the quantity $L = \ln(P/P_0)$, where $P_0 = 1atm$ is a constant. The measured value is $L = 0.6931 \pm 0.0050$.

The gas constant is calculated using the relation $R = \dfrac{PV}{nT} = \dfrac{Ve^L}{nT}$.

Which of the following statements is/are CORRECT? (Use $e^{0.6931} \approx 2.000$).

A) The relative error in the calculated pressure P is approximately 0.50%

B) The percentage error in the measurement of Volume (V) and Temperature (T) contribute equally to the total relative error in R.

C) The absolute error in the calculated gas constant R is approximately $0.0012\,L.atm.K^{-1}.mol^{-1}$.

D) If the uncertainty in L was halved, the total relative error in R would decrease by exactly 50%.

For More Join: @IITJEE_Advanced

29. A point charge +q of mass m is released from rest at the origin $(0,0,0)$ in a vacuum. The region contains a uniform electric field $\vec{E} = E_0 \hat{j}$ and a uniform magnetic field $\vec{B} = B_0 \hat{k}$. The particle undergoes periodic motion in the $xy-$plane. Which of the following statements is/are CORRECT?

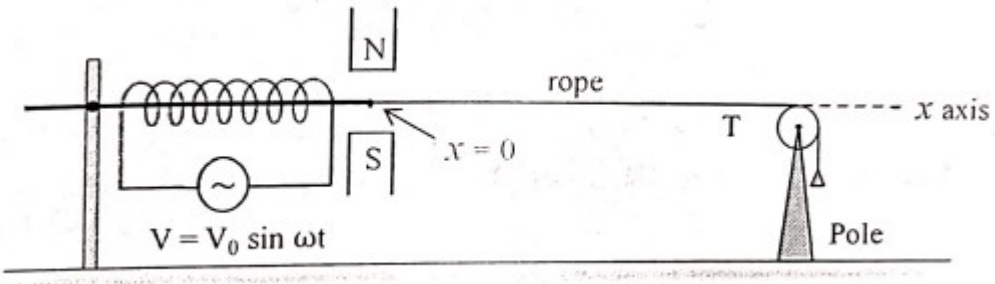
A) The maximum y-coordinate reached by the particle is $y_{max} = \frac{2mE_0}{qB_0^2}$, and the curvature of the path at this point is $\rho = \frac{4mE_0}{qB_0^2}$.

B) The speed of the particle at any height y is given by $\upsilon = \sqrt{\frac{2qE_0 y}{m}}$, and the maximum speed attained is exactly twice the drift velocity $\upsilon_d = \frac{E_0}{B_0}$

C) At the highest point of the trajectory, the magnitude of the net acceleration of the particle is exactly equal to its initial acceleration at the origin.

D) The average velocity of the particle over one full cycle of motion is $\vec{\upsilon}_{avg} = \frac{E_0}{B_0}\hat{i}$, while the net work done by the magnetic force over this cycle is zero.

30. One end of a long and thin rope, stretched horizontally with a tension $T = 8$ N, along x axis, is supporting a weight after passing over a pulley fixed on a vertical pole (see figure). At the other end, a simple harmonic oscillator (a clamped iron rod along the axis of a solenoid fed with AC voltage and oscillating between north and south poles) at $x = 0$, generates a transverse wave of frequency 100 Hz and an amplitude of 2 cm, in the rope. The wave propagates along the rope. The mass per unit length of the rope is 20 g/m. Ignoring the effect of gravity (on the rope), the correct option(s) is /are:



A) Wavelength of the transverse wave is 20 cm.

B) Maximum magnitude of transverse acceleration of any point on the rope is nearly $800\ ms^{-2}$.

C) If the oscillator produces maximum negative displacement at $x = 0$ at time $t = 0$, the equation of the wave can be expressed as $y(x,t) = -0.02\sin[10\pi x - 100\pi t]$ *in SI units.*

D) Tension in the given rope remaining unchanged, if a harmonic oscillator of frequency 200 Hz is used (instead of earlier frequency 100 Hz), the wavelength will be 10 cm.

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SECTION - 3 (Maximum Marks : 24)

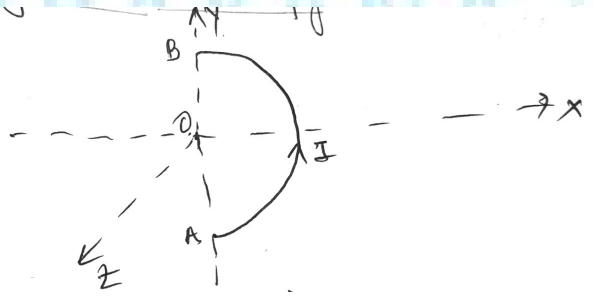
This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

31. Two flat, parallel glass plates are separated by a distance *d*. A wetting liquid (contact angle $\theta = 0^0$) of surface tension T forms a circular bridge between the plates. Let R_1 be the radius of the liquid bridge in the plane of the plates (the distance from the central axis to the meniscus). Let R_2 be the radius of curvature of the meniscus in the plane perpendicular to the plates (where $2R_2 = d$) .Unlike the simplified case where $R_1 << R_2$, here the dimensions are comparable, such that $R_1 = 6R_2$. The total attractive force F between the plates is the sum of the force due to the Laplace pressure difference and the direct vertical component of the surface tension acting along the wetted perimeter. The force is expressed in the form: $F = n.\pi R_1 T$ .Find the value of n

32. A neutral, polarizable atom is placed at a distance *r* form a fixed point charge Q. The external electric field $\vec{E}$ causes the atom's electronic cloud to shift, creating an induced dipole moment $\vec{p}$. The polarizability α of the atom is defined by the relation $\vec{p} = \alpha \vec{E}$. As a result, an attractive force F develops between the point charge and the atom. This force can be expressed in the following form: $F = \dfrac{n.\alpha Q^2}{(4\pi \in_0)^2 r^5}$ Find the integer value of *n*.

33. A monoatomic ideal gas undergoes a polytropic process described by the equation $PV^n =$ constant. In this specific process, the molar heat capacity C of the gas is found to be exactly half of its molar heat capacity at constant volume (C_V). The process starts at a state (P_0, V_0) and ends at a state where the volume is tripled $(3V_0)$. During this expansion, the work done by the gas is W. The heat absorbed by the gas during this same expansion is Q. The ratio of the work done to the heat absorbed is found to be:
$\dfrac{W}{Q} = K$, Find the value of $|K|$.

34. A part of a conductor carrying current I is in the from of a semicircle AB of raduis R and lying in $X-Y$ plane with its center at origin as shown in figure.

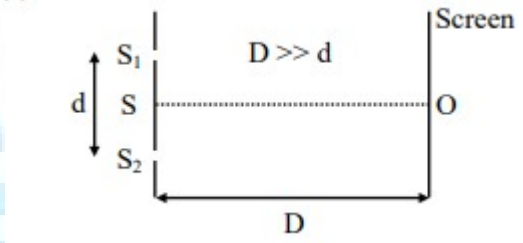


The magnitude of $\oint \vec{B}\,\vec{dl}$ for the circular loop along $3x^2 + 3z^2 = R^2$ in xz- plane due to current is curve AB is $\mu_o KI$. Find K

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35. A coaxial cable consists of an inner solid cylindrical conductor of radius a and a thin out cylindrical shell of radius $b = e^2$ (where e is the base of the natural logarithm). The space between the cylinders is filled with a dielectric material of permittivity ε_0. The dielectric has a maximum "breakdown" electric field strength E_{max}. If the potential difference between the cylinders is increased, the dielectric will first spark at the location where the electric field is strongest. Let V_{max} be the maximum potential difference that can be applied before breakdown occurs. The total electrostatic energy stored per unit length of the cable at this critical moment is given by: $U_{max} = \frac{k.\pi\varepsilon_0 a^2 E_{max}^2}{4}$ Find the value of k.

36. In Young's double slit experiment, a fine beam of coherent monochromatic light of wavelength $\lambda = 600$ nm is incident on identical slits S_1 and S_2 at separation d. The intensity at the central maximum formed at O is I_{max} and the angular fringe width is $\beta = 0.1^\circ$. When a thin transparent film is placed in front of the slit S_2, the intensity at O becomes half the maximum intensity $\left(\text{i.e. } \frac{I_{max}}{2}\right)$, *is* 250 nm. Neglecting the absorption of light by the film, the zero order fringe earlier at O now forms at O' where $OO' = 0.5\,mm$.



Find the refractive index of film.

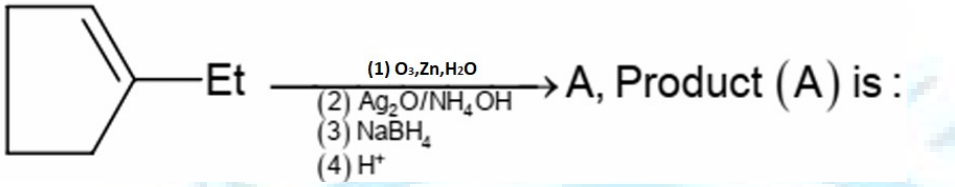
For More Join: @IITJEE_Advanced

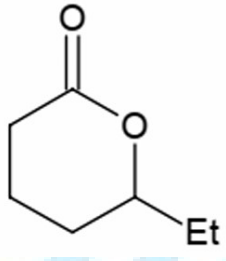
CHEMISTRY **MAX.MARKS: 66**

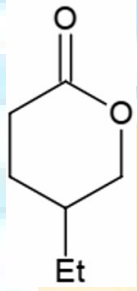
SECTION – 1 (Maximum Marks: 18)

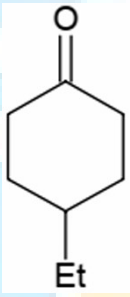
This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

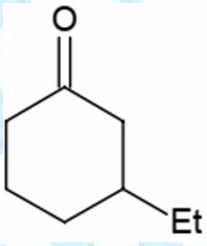
37.



A) 

B) 

C) 

D) 

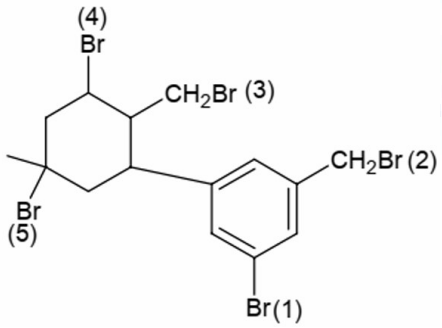
38. Select the group of species in which all show sp^3d hybridization

A) PF_5, IF_5, XeF_4 B) ClO_4^-, IF_7, CO_3^{2-} C) I_3^-, XeF_2, SF_4 D) XeF_6, PF_6^-, lCl_2^+

39. A 1.025 gm sample containing a weak acid HX (mol. wt = 82) is dissolved in 60 ml H_2O and titrated with 0.25 M NaOH solution. When half of the acid is neutralized the pH was found to be 5.0 and the pH at equivalence point is 9.0. Calculate the percentage of HX in the original sample. (Assume law of conservation of volume is obeyed)

A) 40% B) 80% C) 65% D) 78%

40.



Arrange the above mentioned bromine atoms in decreasing order of their reactivity towards hydrolysis through S_N1 path

A) $2>1>5>4>3$ B) $5>2>1>3>4$ C) $2>5>4>3>1$ D) $5>4>2>1>3$

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41. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained $0.1M\ MnO_4^-$ and $0.1M\ H^+$ and which was treated with Fe^{+2} necessary to reduce 90% of $KMnO_4$ to Mn^{+2}. $E^0_{MnO_4^-/Mn^{+2}} = 1.51V$ (Assume that the concentration of H^+ is maintained constant) $(\log 3 = 0.4771)$

A) 1.31 V B) 1.73 V C) 1.38 V D) 1.40 V

42. An inorganic compound 'X' on treatment with concentrated H_2SO_4 produces brown fumes and gives dark brown ring with $FeSO_4$ in presence of concentrated H_2SO_4. Also compound 'X' gives precipitate 'Y', when its solution in dilute HCl is treated with H_2S gas. The precipitate 'Y' on treatment with concentrated HNO_3 followed by excess of NH_4OH further gives deep blue coloured solution, compound 'X' is:

A) $Co(NO_3)_2$ B) $Pb(NO_2)_2$ C) $Cu(NO_3)_2$ D) $Zn(NO_3)_2$

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

43. Select the correct option (s) about real gas

A) If $T = 2T_C$ and $P = 4P_C$ it must be in the gaseous form

B) If $T = T_C/2$ and $P = P_C$ it must be in liquid form

C) If $T = T_C$ and $P = P_C$, then liquid and gaseous form are undistinguishable

D) If $T = 0.9T_C$ and $P > P_C$ then both liquid and gaseous form co-exists

44. Hydrogen fluoride is used in the manufacture of freons (which destroy ozone in the stratosphere) and in the production of aluminium metal. It is prepared by the reaction:

$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$

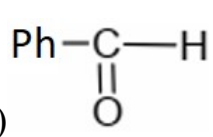
In one process $15.6\,kg$ of CaF_2 are treated with an excess of H_2SO_4 and yield 6kg of HF. Choose the correct statement(s) : [At wt. of Ca = 40 ; S = 32 ; F = 19 ; H = 1]

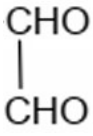
A) % yield of reaction is 75% B) 20.4 kg of $CaSO_4$ is formed

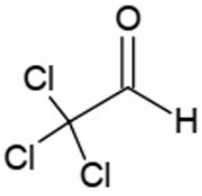
C) 9.8 kg of H_2SO_4 is consumed D) Total mass of products formed is 25.4 kg

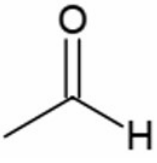
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45. Which of the following compounds will disproportionate on treating with conc. NaOH?

A) 

B) 

C) 

D) 

46. Which of the following is/are correct

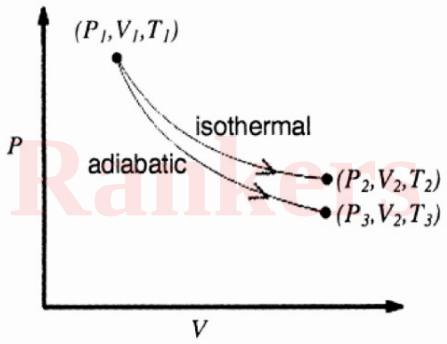
A) $\left[CrCl_2(CN)_2(NH_3)_2\right]$ and $\left[CrCl_3(NH_3)_3\right]$, both have d^2sp^3 hybridization

B) Magnetic moment of $[PdCl_4]^{2-}$ is zero

C) $\left[Cu(NH_3)_4\right]^{2+}$, $\left[Pt(NH_3)_4\right]^{2+}$ and $\left[Ni(CN)_4\right]^{2-}$, all have dsp^2 hybridization of central metal

D) It is difficult to substitute chelating ligands compared to similar monodentate ligands

47. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) incorrect



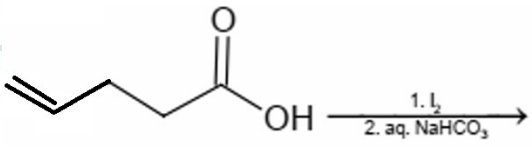
A) $T_1 = T_2$

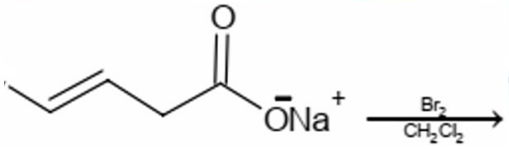
B) $T_3 > T_1$

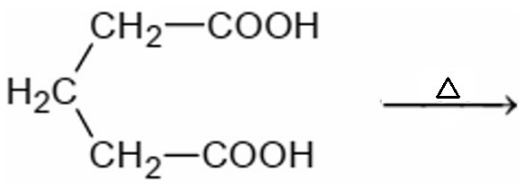
C) $|w_{isothermal}| < |w_{adiabatic}|$

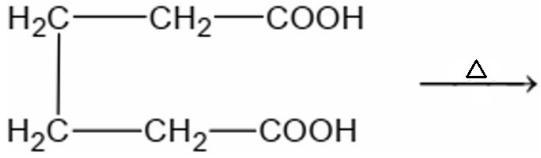
D) $\Delta U_{isothermal} > \Delta U_{adiabatic}$

48. Which of the following reaction (s) can generate a five membered ring as the major product?

A) 

B) 

C) 

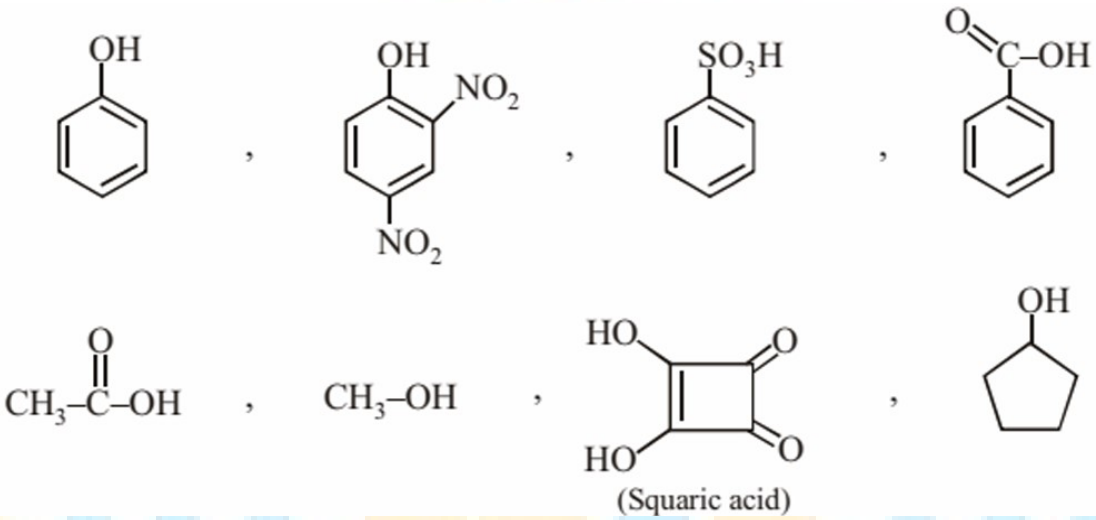
D) 

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SECTION - 3 (Maximum Marks : 24)

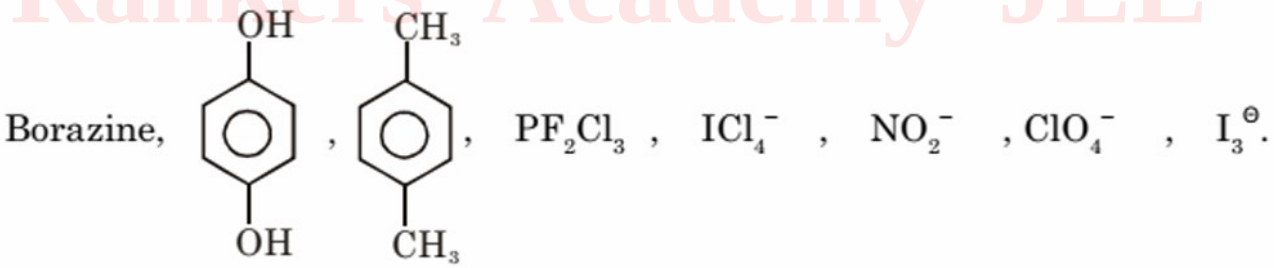
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Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

49. How many compound(s) can release efferevescence of CO_2 when react with $NaHCO_3$:



(Squaric acid)

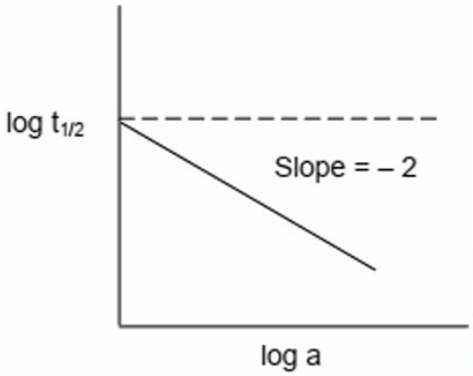
50. How many of the given species is/are polar.



51. A compound (A) having molecular formula $C_7H_{11}Br$ is optically active. A reacts with HBr in the absence of peroxide to give isomeric products (B) and (C). Treating (A) with potassium-t-butoxide gives (D). (D) on reductive ozonolysis gives two moles of formaldehyde and one mole of 1,3 cyclopentane dione. (A) in the presence of peroxide reacts with HBr to give (E). The total number of chiral carbons in one molecule of (E) is:

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52. A graph between $\log t_{1/2}$ and $\log a$ (abscissa), 'a' being the initial concentration of A in the reaction $A \rightarrow$ product is depicted in figure.



The rate law is $-\frac{d[A]}{dt} = k[A]^n$

The value of n will be

53. Equilibrium constant (K) at $27^0 C$ for the decomposition of poisonous gas $COCl_2$, represented by $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ is $\frac{1}{2} \times 10^{-7}$ bar

Find magnitude of $\Delta_r G$ (kJ/mole) for the reaction $\left(CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)\right)$ at $27^0 C$. Given that mixture of 1 mole $COCl_2(g)$, 1 mole $CO(g)$ and 2 mole $Cl_2(g)$ is taken in closed container and maintained at 1 bar pressure and $27^0 C$. (R = 8.3 J/mole/K, $\log 2 = 0.3$)

54. Decomposition of non-volatile solute 'A' into another non-volatile solute B and C, when dissolved in water follows first order kinetics as:

$A \rightarrow 2B + C$

When 2 moles of A is dissolved in 180 gm of water and left for decomposition, the vapour pressure of solution was found to be 20 mm Hg after 12 hrs. Assume constant temperature of $25^0 C$, throughout. The vapour pressure of pure water at $25^0 C$ is 26 mm Hg. Then determine the vapour pressure (in mmHg) of the solution after 24 hrs.

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